

Martin Birkenbaum

SENIOR SOFTWARE ENGINEER

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Senior Software Engineer with 8+ years of experience designing, building, and deploying large-scale distributed systems, microservices, and advanced automation, including ML infrastructure components. Passionate about robust testing, efficient developer toolchains, and clean code.

SKILLS

Languages:	Python, TypeScript, JavaScript, SQL, HTML/CSS
Technologies & Frameworks:	Docker, Kubernetes, Git, GitOps, PostgreSQL, Jinja2, WeasyPrint, pytest, Distributed Systems, Data Structures, Algorithms, Machine Learning Infrastructure, Natural Language Processing (NLP)

WORK EXPERIENCE

Senior Software Engineer at **Initech Corporation** 2021-03-01 – 2026-07-01

Led architecture and development of high-throughput microservices and GitOps-driven infrastructure pipelines.

- Designed and implemented high-throughput Python-based event-driven microservices, processing 10M+ daily events within large-scale distributed systems, supporting critical data processing needs.
- Standardized scalable containerized deployments utilizing Docker, Kubernetes, and GitOps workflows, contributing to robust ML infrastructure practices.
- Mentored 6 junior/mid-level engineers, fostering best practices in code quality, styling, and rigorous testing within a collaborative environment.
- Developed custom internal toolchains that optimized CI pipeline execution times by 40%, enhancing development efficiency and debugging capabilities.

Software Engineer at **Hooli Inc.** 2018-06-01 – 2021-02-01

Backend engineer focused on building RESTful APIs, developer tooling, and automated deployment integrations.

- Developed robust REST APIs in Python/Django and TypeScript/Node.js, focusing on performance and scalability.
- Integrated third-party APIs and automated webhook processors for external developer platforms, ensuring seamless data exchange and system interoperability.
- Implemented a suite of unit and integration tests, raising code coverage from 60% to 92% and reinforcing a commitment to high-quality software.

PROJECTS

JobGitOps (Python, Nix, GitHub Actions) 2026-06-01 – 2026-07-01

A GitOps-driven job application and tracking system utilizing GitHub Issues, Projects, and Actions.

- Developed python-jobspy scraper integration and two-pass LLM fit triage
- Designed reproducible local environments via Nix and devenv

EDUCATION

Bachelor of Science in Computer Science at **University of Washington**

2014-09-01 – 2018-06-01

GPA: 3.8